START

PARAMETERS

$\vec{R}_0,\ t_0,\ \vec{v}_0,\ N,\ \Delta t$

INPUT

$\vec{R}(t)$ ← POSITION OF PLANETS

ΔT ← ΔT FOR PLANETS

M_PLANETS ← MASS OF ALL PLANETS

M_* ← MASS OF STAR

WE TAKE IN INITAL POSITION, VELOCITY, AND TIME. WE ALSO TAKE IN Δt, AND N, WHICH IS AMOUNT OF STEPS Δt. (NOTE THAT Δt (ROCKET TIME) AND ΔT (PLANET TIME) ARE SEPARATE! WE HAVE $\Delta T = n\Delta t$)

(A)

$$N_k = \frac{\Delta T}{\Delta t}$$

THESE ARRAYS WILL HOLD ALL POSITIONS, VELOCITIES AND ACCELERATIONS FOR THE ROCKET.

$$\vec{R} = \text{ARRAY}(N, 2)$$
$$\vec{v} = \text{ARRAY}(N, 2)$$
$$\vec{a} = \text{ARRAY}(N, 2)$$

SETTING INITIAL CONDITIONS

$$\vec{R}_0 = \vec{R}_0$$
$$\vec{v}_0 = \vec{v}_0$$
$$\vec{a}_0 = G_0$$

(B)

FOR ALL K < N DO

(C) UPDATING ARRAYS

END OF LOOP

END

(A)

FOR THIS PROGRAM WE USE A CLOCK STRUCTURE, WHICH HELPS US WITH INTERPOLATION. WE HAVE OUR ITERATIVE VARIABLE K, WHERE $\Delta t \cdot K = T$ (UP UNTIL NOW). SINCE WE ALSO WANT TO KEEP TRACK OF HOW MANY ΔTs HAVE PASSED, WE DEFINE N_K, WHICH IS HOW MANY TIMES Δt GOES INTO ΔT. WE CAN THEN GET THE CLOSEST MULTIPLE OF ΔT, BY TAKING $K - i$, WHERE $K \equiv_{N_k} i$

MODULO

$t = (K - i)\Delta t$ $\quad$ $t = K \cdot \Delta t$ $\quad$ $t = (K - i + 1)\Delta t$

THESE TWO POINTS CAN BE REACHED USING MULTIPLES OF ΔT (PLANET TIME)

THIS ONE CANNOT, AND WE MUST INTERPOLATE

(B)

HERE WE CALCULATE THE GRAVITATIONAL ACCELERATION ON THE VESSEL

$$a_x = 0$$
$$a_y = 0$$

$$t_1 = K - i$$
$$t_2 = t_1 + 1$$

← SE (A) FOR EXPLANATION

$$I = \frac{i}{N_k}$$

← THIS IS OUR INTERPOLATION VALUE, $I \in [0, 1)$, WHICH GETS CLOSER TO 0 AND 1 WHEN K IS CLOSE TO A MULTIPLE OF ΔT

NUM_PLANETS IS ALWAYS 7

FOR all j < NUM_PLANETS DO

$$\vec{R}_{j_1} = \vec{R}(t_1)$$
$$\vec{R}_{j_2} = \vec{R}(t_2)$$

$$\Delta R_{j_x} = R_{j_{2_x}} - R_{j_{1_x}}$$
$$\Delta R_{j_y} = R_{j_{2_y}} - R_{j_{1_y}}$$

$$R_{j_x} = R_{j_{1_x}} + \Delta R_{j_x} \cdot I$$
$$R_{j_y} = R_{j_{1_y}} + \Delta R_{j_y} \cdot I$$

← $\vec{R}_j$ IS THE INTERPOLATED POSITION OF PLANET j

$$r_{j_x} = R_{k_x} - R_{j_x}$$
$$r_{j_y} = R_{k_y} - R_{j_y}$$

← $\vec{R}_k$ IS THE CURRENT POSITION OF THE VESSEL

G IS THE GRAVITATIONAL CONSTANT

$$\gamma = G \frac{\text{M\_PLANETS}_j}{(r_{j_x}^2 + r_{j_y}^2)^{\frac{3}{2}}}$$

$$a_x \mathrel{+}= \gamma\, r_{j_x}$$
$$a_y \mathrel{+}= \gamma\, r_{j_y}$$

END OF LOOP

OUT a_x, a_y

(C)

$$R_{k+1_x} = R_{k_x} + V_{k_x} \cdot \Delta t + \frac{1}{2} a_{k_x} \cdot \Delta t^2$$
$$R_{k+1_y} = R_{k_y} + V_{k_y} \cdot \Delta t + \frac{1}{2} a_{k_y} \cdot \Delta t^2$$

$$a_{k+1} = G_{k+1}$$

(B) NOTE THAT INPUT (K WITHIN G) IS NOTED IN SUBSCRIPT

$$v_{k+1_x} = v_{k_x} + \frac{1}{2}\left(a_{k_x} + a_{k+1_x}\right)\Delta t$$
$$v_{k+1_y} = v_{k_y} + \frac{1}{2}\left(a_{k_y} + a_{k+1_y}\right)\Delta t$$